

# The Swiss Franc Shock and Polish Voter Attitudes: Does Information Increase Support for Government Intervention?

Session 3 Analysis Report

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## Abstract

In January 2015 the Swiss National Bank (SNB) unexpectedly abandoned its currency floor, sending the Swiss franc sharply higher against the Polish zloty and raising monthly payments for the many Polish households holding franc-denominated mortgages. Using a nationally representative survey of 2,044 Polish adults fielded in October 2015 with an embedded randomized information treatment, we ask whether providing information about the crisis increases support for government intervention to help affected borrowers. We merge the survey with administrative demographics, estimate weighted logistic regressions of intervention support on the randomized treatments, and subject the baseline estimate to a battery of robustness checks. Factual information about the franc shock raises support for intervention by a statistically significant margin (coefficient 0.302,  $p = 0.016$ ), while historical-context and Hungary-comparison framings do not. Direct exposure to a current foreign-currency (FX) loan is a far stronger predictor of support than any framing. The information effect is robust to weighting and to the addition of demographic controls on the full sample, but loses conventional significance once income and ideology controls shrink the analytic sample by roughly 44% — a missing-data artifact rather than confounding. We conclude that information shifts attitudes on the margin, but that personal economic exposure dominates, and that sample composition, not the controls themselves, drives the apparent fragility of the effect.

## 1 Introduction

The 15 January 2015 removal of the SNB’s CHF/EUR floor was a textbook external economic shock: unanticipated, large, and unevenly distributed across the population. Figure 1 shows the resulting discontinuity in the CHF/PLN exchange rate, which fed directly into the monthly payments of Polish households with franc-denominated mortgages. This report studies the political consequences of that shock using survey data described in the course codebook, drawn from Ahlquist, Copelovitch, and Walter, “The Political Consequences of External Economic Shocks,” *American Journal of Political Science* (2020).

The research question is whether *providing information* about the crisis increases support for government intervention to help affected borrowers, and whether a respondent’s own FX-loan exposure moderates that effect. Because the information framing was randomly assigned, the treatment contrasts have a causal interpretation within the surveyed population.

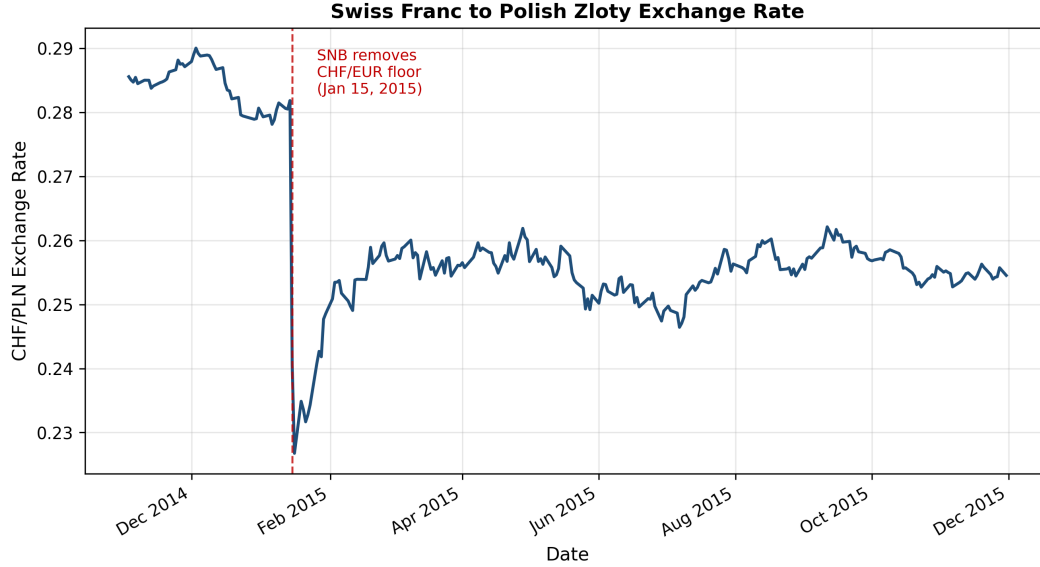


Figure 1: Swiss franc to Polish zloty exchange rate around the SNB’s removal of the CHF/EUR floor on 15 January 2015. The series exhibits a sharp break at the announcement before partially stabilizing over the remainder of 2015.

## 2 Data and Methods

### 2.1 Data sources and merge

Two datasets underlie the analysis. The first, `swiss_franc_survey.csv`, is a survey of 2,044 Polish adults fielded 7–21 October 2015, including the randomized treatment, the intervention-support outcome, FX-loan status, basic demographics, and survey weights. The second, `respondent_demographics.csv`, supplies household income quintile and left–right ideological self-placement.

The demographics file contains 2,994 rows — 1,994 genuine respondents plus roughly 1,000 identifiers that match no one in the survey — and 50 genuine survey respondents are absent from it. Only a **left join** (`survey ← demographics`) preserves the analytic frame: it returns exactly 2,044 rows, assigning missing demographics to the 50 unmatched respondents while discarding the spurious identifiers. Inner, right, and outer joins each distort the sample. We verified the merge by confirming the 2,044-row count, the absence of duplicate respondent IDs, and the expected treatment-arm counts (Control 480, Info 549, History 520, Hungary 495).

### 2.2 Variables and sample

The outcome `supports_intervention` is a binary indicator equal to one if the respondent favors “some” or “big” government intervention; 8 respondents with a “don’t know” or missing response are excluded, leaving 2,036 with a valid outcome. The treatment is the four-arm randomized framing (control, factual information, historical context, Hungary comparison), with the *control* group as the reference category. FX exposure distinguishes current borrowers (69), past borrowers (55), and those who never held an FX loan (1,918). Following the codebook, an income quintile of 0 (“no income”) is treated as a valid category rather than missing. Table 1 reports descriptive statistics for the merged analytic sample.

Table 1: Summary statistics, merged analytic sample.

| Variable                    | N    | Mean  | SD    | Min   | Max   |
|-----------------------------|------|-------|-------|-------|-------|
| Supports intervention (0/1) | 2036 | 0.48  | 0.50  | 0.00  | 1.00  |
| Age (years)                 | 2044 | 48.81 | 17.58 | 18.00 | 85.00 |
| Female (0/1)                | 2044 | 0.54  | 0.50  | 0.00  | 1.00  |
| Education level (1–5)       | 2044 | 2.52  | 1.07  | 1.00  | 4.00  |
| Urban–rural (1–5)           | 2044 | 1.85  | 0.80  | 1.00  | 3.00  |
| Income quintile (1–5)       | 1479 | 3.13  | 1.45  | 0.00  | 5.00  |
| Left–right placement (0–10) | 1502 | 0.47  | 1.56  | –3.00 | 3.00  |
| FX exposed, current (0/1)   | 2044 | 0.03  | 0.18  | 0.00  | 1.00  |
| FX exposed, past (0/1)      | 2044 | 0.03  | 0.16  | 0.00  | 1.00  |

Income and ideology are substantially missing by design — roughly a quarter of demographic records lack each — as documented in the source file (Table A) and detailed in the appendix. This missingness is the central methodological issue for the analysis below.

## 2.3 Estimation

We estimate logistic regressions of `supports_intervention` on the treatment indicators, applying survey weights. The baseline model includes the treatment arms only; subsequent specifications add FX-exposure indicators, demographic controls (age, gender, education, urban–rural settlement), and the income and ideology controls. We track the analytic  $N$  at every step, because adding the partially missing demographic controls mechanically reduces the sample through listwise deletion. Robustness specifications vary the use of survey weights, the control set, the treatment parameterization (separate arms versus a pooled “any information” dummy), and the sample (full versus urban only).

# 3 Results

## 3.1 Main estimates

Table 2 reports the baseline and controls-augmented weighted logistic regressions. The factual information treatment raises support for intervention relative to the control group (0.302,  $p = 0.016$  in the baseline), and the estimate is stable when FX exposure and demographic controls are added on the full sample (0.324). Neither the historical-context nor the Hungary treatment produces a statistically significant effect. By far the largest coefficient belongs to *current* FX exposure (1.287,  $p < 0.01$ ): respondents personally repaying a franc loan are dramatically more supportive of intervention, independent of any framing they received.

Table 2: Logistic regression of support for government intervention (weighted).

|                      | (1)<br><i>Baseline</i> | (2)<br><i>+ Controls</i> |
|----------------------|------------------------|--------------------------|
| Info treatment       | 0.302**<br>(0.126)     | 0.324**<br>(0.127)       |
| History treatment    | 0.123<br>(0.128)       | 0.133<br>(0.130)         |
| Hungary treatment    | 0.194<br>(0.130)       | 0.185<br>(0.131)         |
| FX exposed (current) |                        | 1.287***<br>(0.283)      |
| FX exposed (past)    |                        | −0.163<br>(0.289)        |
| Demographic controls | No                     | Yes                      |
| Survey weights       | Yes                    | Yes                      |
| N                    | 2036                   | 2035                     |

\*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ .

### 3.2 Robustness

Table 3 arrays six specifications side by side; the compiled version is reproduced in Figure 2. The information effect survives dropping the survey weights (column 2, 0.290,  $p = 0.021$ ) and adding demographic controls on the full sample (column 3, 0.324,  $p = 0.011$ ). Pooling the three framings into a single “any information” dummy yields a smaller but still significant effect (column 5, 0.218,  $p = 0.042$ ), and restricting to urban respondents leaves the estimate essentially unchanged though borderline (column 6, 0.326,  $p = 0.051$ ).

The one specification in which the effect loses significance is column 4, which adds the income and ideology controls. Here the coefficient remains positive and comparable in magnitude (0.267) but  $p$  rises to 0.118. The reason is sample composition rather than confounding: listwise deletion of missing income and ideology drops the analytic sample from 2,035 to 1,151 — about 44% — and the loss of precision, not a change in the point estimate, drives the result. Across every specification, current FX exposure remains large and highly significant.

Table 3: Logistic Regression: Support for Government Intervention

|                      | (1)                | (2)                | (3)                 | (4)                 | (5)                 | (6)                 |
|----------------------|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
|                      | Base (weighted)    | Base (unweighted)  | + Controls          | Full controls       | Pooled treatment    | Urban only          |
| Info treatment       | 0.302**<br>(0.126) | 0.290**<br>(0.126) | 0.324**<br>(0.127)  | 0.267<br>(0.170)    |                     | 0.326*<br>(0.167)   |
| History treatment    | 0.123<br>(0.128)   | 0.047<br>(0.127)   | 0.133<br>(0.130)    | 0.261<br>(0.174)    |                     | 0.242<br>(0.169)    |
| Hungary treatment    | 0.194<br>(0.130)   | 0.145<br>(0.129)   | 0.185<br>(0.131)    | 0.225<br>(0.176)    |                     | 0.221<br>(0.169)    |
| Any information      |                    |                    |                     |                     | 0.218**<br>(0.107)  |                     |
| FX exposed (current) |                    |                    | 1.287***<br>(0.283) | 1.251***<br>(0.352) | 1.290***<br>(0.283) | 1.256***<br>(0.319) |
| FX exposed (past)    |                    |                    | -0.163<br>(0.289)   | -0.168<br>(0.356)   | -0.158<br>(0.289)   | -0.049<br>(0.331)   |
| Demographic controls | No                 | No                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Survey weights       | Yes                | No                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Sample               | Full               | Full               | Full                | Full                | Full                | Urban               |
| N                    | 2036               | 2036               | 2035                | 1151                | 2035                | 1208                |

\*\*\*  $p < 0.01$ ; \*\*  $p < 0.05$ ; \*  $p < 0.1$ . Reference: control group, no FX loan.

| Logistic Regression: Support for Government Intervention |                  |                   |                   |                   |                   |                   |
|----------------------------------------------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                                          | (1)              | (2)               | (3)               | (4)               | (5)               | (6)               |
|                                                          | Base (weighted)  | Base (unweighted) | + Controls        | Full controls     | Pooled treatment  | Urban only        |
| Info treatment                                           | 0.302 * *        | 0.290 * *         | 0.324 * *         | 0.267<br>(0.170)  |                   | 0.326*<br>(0.167) |
| History treatment                                        | 0.123<br>(0.128) | 0.047<br>(0.127)  | 0.133<br>(0.130)  | 0.261<br>(0.174)  |                   | 0.242<br>(0.169)  |
| Hungary treatment                                        | 0.194<br>(0.130) | 0.145<br>(0.129)  | 0.185<br>(0.131)  | 0.225<br>(0.176)  |                   | 0.221<br>(0.169)  |
| Any information                                          |                  |                   |                   |                   | 0.218 * *         |                   |
| FX exposed (current)                                     |                  |                   | 1.287 * *         | 1.251 * *         | 1.290 * *         | 1.256 * *         |
| FX exposed (past)                                        |                  |                   | -0.163<br>(0.289) | -0.168<br>(0.356) | -0.158<br>(0.289) | -0.049<br>(0.331) |
| Demographic controls                                     | No               | No                | Yes               | Yes               | Yes               | Yes               |
| Survey weights                                           | Yes              | No                | Yes               | Yes               | Yes               | Yes               |
| Sample                                                   | Full             | Full              | Full              | Full              | Full              | Urban             |
| N                                                        | 2036             | 2036              | 2035              | 1151              | 2035              | 1208              |

\*\*\*  $p < 0.01$ ; \*\*  $p < 0.05$ ; \*  $p < 0.1$ . Reference: control group, no FX loan.

Figure 2: Compiled robustness table (rendered from `robustness_table.tex`).

## 4 Conclusion

Factual information about the Swiss franc shock measurably increased Polish respondents' support for government intervention to help affected borrowers, whereas historical and comparative framings

did not. The effect is modest but robust to weighting, to control specification on the full sample, and to a pooled treatment definition. Its apparent fragility under the fullest control set is a missing-data artifact — a 44% sample reduction from listwise deletion of income and ideology — rather than evidence that the controls explain the effect away. The most substantively important finding is that personal economic exposure dominates information: respondents currently repaying a franc loan were far more supportive of intervention than any framing could induce. For applied work, the analysis is a reminder to predict and report how the analytic sample changes as controls are added, and to distinguish a vanishing point estimate from a vanishing standard error.

## A Appendix: Demographic Distributions

The demographics file is summarized below over all 2,994 rows (Table A), with the full income-quintile and ideology distributions in Tables A and A and the corresponding figures following. These tables document the missingness in income and ideology that drives the sample reduction in column 4 of the robustness table.

Table 4: Respondent demographics summary (`respondent_demographics.csv`, all rows)

| Statistic                  | Value         |
|----------------------------|---------------|
| Total rows                 | 2,994         |
| <i>Income quintile</i>     |               |
| Non-missing                | 2,215 (74.0%) |
| Missing                    | 779 (26.0%)   |
| Modal category             | Q5            |
| <i>Left-right ideology</i> |               |
| Non-missing                | 2,257 (75.4%) |
| Missing                    | 737 (24.6%)   |
| Mean                       | 0.49          |
| Median                     | 0.0           |
| Std. dev.                  | 1.58          |

Table 5: Respondent income quintile distribution

| Income quintile | Count | % of valid | % of all |
|-----------------|-------|------------|----------|
| No income       | 11    | 0.5        | 0.4      |
| Q1              | 444   | 20.0       | 14.8     |
| Q2              | 339   | 15.3       | 11.3     |
| Q3              | 495   | 22.3       | 16.5     |
| Q4              | 395   | 17.8       | 13.2     |
| Q5              | 531   | 24.0       | 17.7     |
| Missing         | 779   |            | 26.0     |
| Total           | 2,994 | 100.0      | 100.0    |

Table 6: Respondent left–right ideology distribution

| Left–right | Count | % of valid | % of all |
|------------|-------|------------|----------|
| -3         | 95    | 4.2        | 3.2      |
| -2         | 125   | 5.5        | 4.2      |
| -1         | 242   | 10.7       | 8.1      |
| 0          | 894   | 39.6       | 29.9     |
| +1         | 303   | 13.4       | 10.1     |
| +2         | 221   | 9.8        | 7.4      |
| +3         | 377   | 16.7       | 12.6     |
| Missing    | 737   |            | 24.6     |
| Total      | 2,994 | 100.0      | 100.0    |

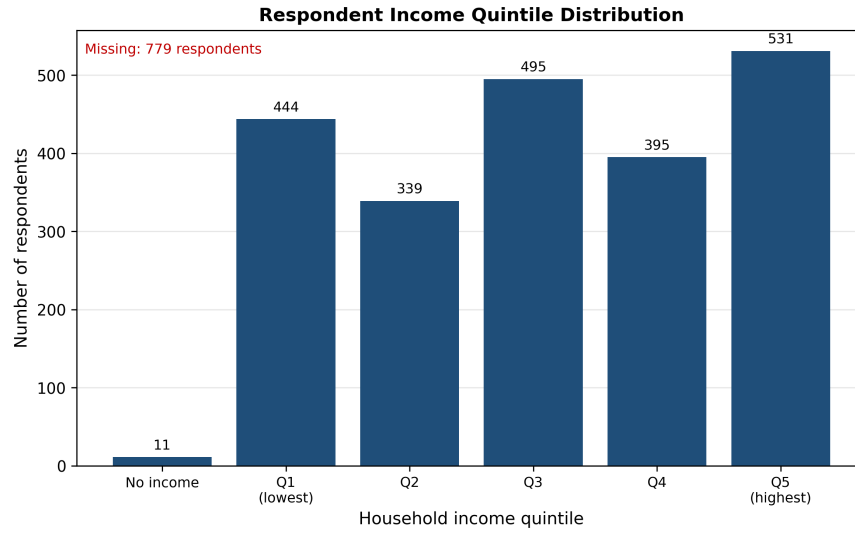


Figure 3: Distribution of household income quintile.

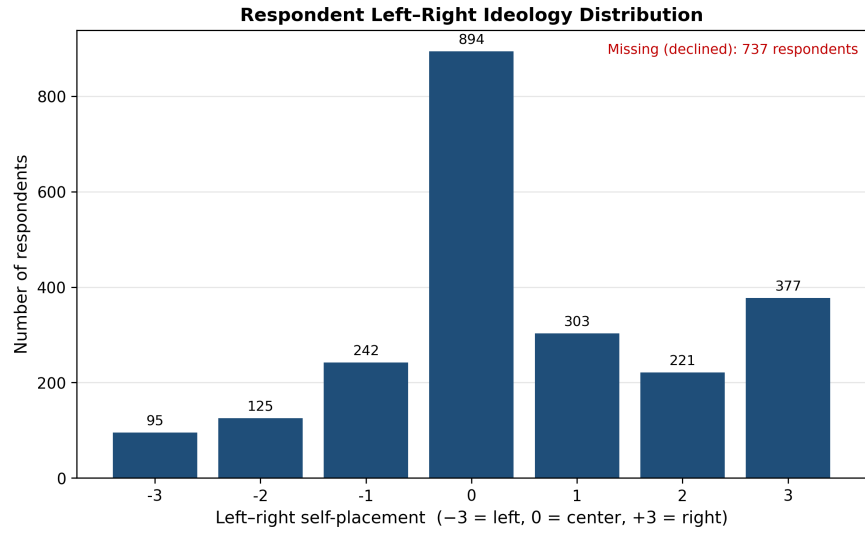


Figure 4: Distribution of left-right ideological self-placement.

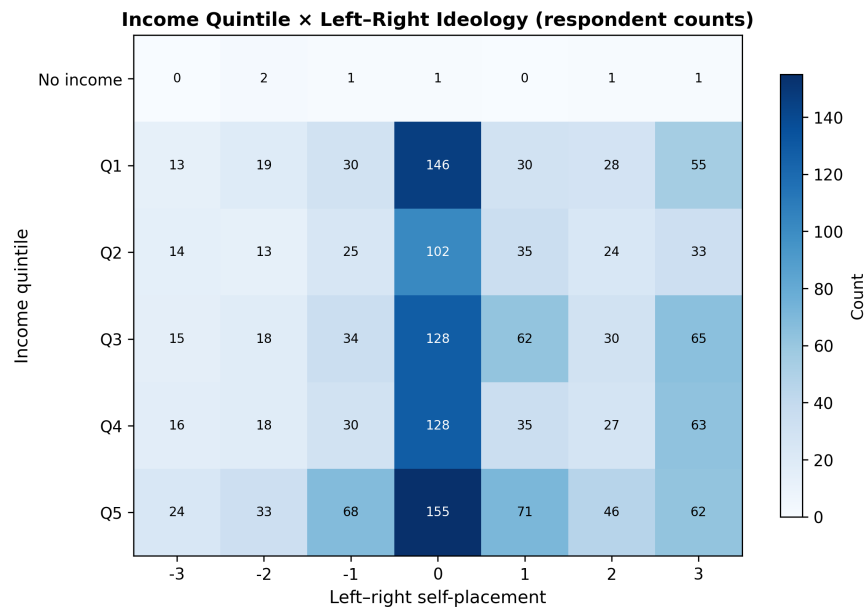


Figure 5: Cross-tabulation of income quintile and left-right ideology.